

Functions and Function Notation

ANSWER KEY



Function notation

1 Consider the function $f(x) = 2x^2 - 5x + 3$. Evaluate:

a $f(0) = 3$

b $f(2) = 1$

c $f(-1) = 10$

d $f\left(\frac{1}{2}\right) = 1$

2 Consider the function $g(x) = \sqrt{x+5}$. Evaluate:

a $g(4) = 3$

b $g(-1) = 2$

c $g(11) = 4$

d $g(-5) = 0$

3 Consider the function $h(x) = \frac{x-3}{x+2}$. Evaluate:

a $h(0) = -\frac{3}{2}$

b $h(3) = 0$

c $h(-1) = -4$

d $h(5) = \frac{2}{7}$

4 If $f(x) = 3x - 7$ and $f(a) = 11$, find the value of a .

$3a - 7 = 11$, so $3a = 18$ and $a = 6$.

Identifying functions

5 Determine whether each equation defines y as a function of x :

a $y = 2x + 1$ Yes — one output for each input.

b $x = y^2$ No — e.g. $x = 4$ gives $y = 2$ or $y = -2$.

c $y = |x|$ Yes — one output for each input.

d $x^2 + y^2 = 9$ No — a circle fails the vertical line test.

6 Explain how the vertical line test is used to decide whether a graph represents a function.

If every vertical line crosses the graph at most once, each input x has at most one output y , so the graph represents a function. If any vertical line crosses the graph more than once, it is not a function.

Combining functions

7 Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find:

a $(f+g)(x) = x^2 + 2x - 2$

b $(f-g)(x) = x^2 - 2x + 4$

c $(f \cdot g)(2) = 5 \cdot 1 = 5$

d $(f \circ g)(1) = f(-1) = 2$

- 8 A tutoring center charges a one-time registration fee plus an hourly rate, modeled by $C(n) = 250 + 45n$ riyals for n hours of tutoring.
- a Interpret the value $C(0)$ in context. $C(0) = 250$: the registration fee alone, before any tutoring hours.
 - b Find $C(12)$. $C(12) = 250 + 540 = 790$ riyals.
 - c Solve $C(n) = 925$ and interpret your answer. $45n = 675$, so $n = 15$: a total cost of 925 riyals buys 15 hours of tutoring.
- 9 Let $f(x) = \sqrt{x}$ and $g(x) = x - 4$. Find the domain of $(f \circ g)(x)$.
 $(f \circ g)(x) = \sqrt{x - 4}$, which requires $x - 4 \geq 0$: domain $x \geq 4$.

Piecewise-defined functions

- 10 A function is defined by $f(x) = x + 4$ for $x < 1$, and $f(x) = 2x + 1$ for $x \geq 1$. Evaluate:
- a $f(-3) = 1$
 - b $f(0) = 4$
 - c $f(1) = 3$
 - d $f(4) = 9$
- 11 A phone plan charges a flat 60 riyals for up to 5 GB of data, then 15 riyals per additional GB beyond 5 GB.
- a Write a piecewise formula $C(x)$ for the monthly cost, where x is data used in GB. $C(x) = 60$ for $0 \leq x \leq 5$, and $C(x) = 60 + 15(x - 5)$ for $x > 5$.
 - b Find $C(3)$ and $C(8)$. $C(3) = 60$ riyals; $C(8) = 60 + 15(3) = 105$ riyals.
 - c Is C continuous at $x = 5$? Explain. Yes — both pieces give $C(5) = 60$, so there is no jump at $x = 5$.
- 12 Sketch-reasoning: the graph of $f(x) = |x - 2|$ can be written as a piecewise function without absolute value bars. Write it as such, and state $f(-1)$ and $f(5)$.
 $f(x) = -(x - 2) = 2 - x$ for $x < 2$, and $f(x) = x - 2$ for $x \geq 2$. $f(-1) = 3$; $f(5) = 3$.

Even, odd, and one-to-one functions

- 13 Determine whether each function is even, odd, or neither:
- a $f(x) = x^4 - 3x^2$ Even — $f(-x) = f(x)$.
 - b $f(x) = x^3 - x$ Odd — $f(-x) = -f(x)$.
 - c $f(x) = x^2 + 2x$ Neither — matches neither pattern (e.g. $f(-1) = -1$, $f(1) = 3$, and $-f(1) = -3$).
- 14 Determine whether each function is one-to-one (passes the horizontal line test). If not, give a horizontal line that crosses the graph twice.
- a $f(x) = 2x + 5$ One-to-one — every horizontal line meets the graph at most once.
 - b $f(x) = x^2$ Not one-to-one — e.g. $y = 4$ meets the graph at $x = \pm 2$.

Reading function graphs

- 15** The graph of $f(x) = x^2 - 2$ is shown. Use it to estimate $f(1)$, and state the interval(s) where f is increasing.

From the graph, $f(1) = 1^2 - 2 = -1$. The parabola is increasing for $x > 0$.

- 16** Using the graph of $f(x) = x^2 - 2$ above, estimate all x -values where $f(x) = 2$.

Setting $x^2 - 2 = 2$: $x^2 = 4 \Rightarrow x = \pm 2$, matching where the parabola crosses the horizontal line $y = 2$ on the graph.

Average rate of change from function notation

- 17** For $f(x) = x^2 + 1$, find the average rate of change of f from $x = 1$ to $x = 3$.

$$\frac{f(3) - f(1)}{3 - 1} = \frac{10 - 2}{2} = 4.$$

- 18** A ball's height is modeled by $h(t) = -5t^2 + 20t$ (meters, seconds). Find the average rate of change of height from $t = 1$ to $t = 2$, and interpret the sign of your answer.

$\frac{h(2) - h(1)}{2 - 1} = \frac{20 - 15}{1} = 5$. Since the average rate of change is positive, the ball's height is increasing on average over this interval.

Difference quotients

- 19** For $f(x) = x^2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$.

$$\frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h.$$

- 20** For $f(x) = 3x + 2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$, and explain why the result makes sense for a linear function.

$\frac{3(x+h) + 2 - (3x + 2)}{h} = \frac{3h}{h} = 3$. This makes sense because a linear function has constant slope 3 everywhere, so its average rate of change over any interval is always exactly that slope.

Inverse functions (introduction)

- 21** Find the inverse of $f(x) = 3x - 4$, and verify by checking $f(f^{-1}(x)) = x$.

Swap and solve: $x = 3y - 4 \Rightarrow y = \frac{x+4}{3}$. So $f^{-1}(x) = \frac{x+4}{3}$. Check: $f\left(\frac{x+4}{3}\right) = 3\left(\frac{x+4}{3}\right) - 4 = x + 4 - 4 = x$ ✓.

- 22** Explain why $f(x) = x^2$ (with domain all reals) does not have an inverse function, but $f(x) = x^2$ restricted to $x \geq 0$ does.

$f(x) = x^2$ is not one-to-one over all reals (e.g. $f(2) = f(-2) = 4$), so it fails the horizontal line test and has no inverse function. Restricting to $x \geq 0$ makes it one-to-one, so an inverse ($f^{-1}(x) = \sqrt{x}$) exists.